Here are **very detailed, easy, exam-oriented notes** for **Networking – Unit 4: Data Link Layer & Protocols**.
I've written them in **simple language + examples + diagrams (ASCII) + bullet points** to make it fully understandable and perfect for your college exams.

UNIT 4 – DATA LINK LAYER (Detailed Notes)

Topics Covered

- Error Detection Techniques
- Error Correction Techniques
- Data Link Control
- Framing
- Flow Control
- Error Recovery Protocols
- Stop-and-Wait ARQ
- Go-Back-N ARQ
- Point-to-Point Protocol (PPP)

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1. DATA LINK LAYER (DLL)

Definition

The Data Link Layer is the **second layer** of the OSI model.
It provides **reliable node-to-node delivery** of data over a physical channel.

Main Responsibilities

1. **Framing**
2. **Error Detection & Correction**
3. **Flow Control**
4. **Error Control (ARQ Protocols)**
5. **Access Control (MAC)**

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2. ERROR DETECTION TECHNIQUES

Error detection helps the receiver know **whether the data is corrupted** during transmission.

2.1 Parity Check

- Adds **1 extra bit (parity bit)** to the data.

- Two types:
    - **Even parity:** Total number of 1s = even
    - **Odd parity:** Total number of 1s = odd
- **Detects only single-bit errors.**
- **Cannot detect burst errors.**

2.2 Checksum

- Data is divided into **equal-sized blocks**.
- All blocks are added using **1's complement addition**.
- The result (sum) is inverted → **Checksum**.
- Receiver adds all blocks + checksum →
  result must be **all 1s** → no error.

2.3 Cyclic Redundancy Check (CRC)

Most powerful error detection method.

Steps:

1. Sender divides data by a **generator polynomial (G)**.
2. Remainder = **CRC bits**.
3. Sender appends CRC bits to frame.
4. Receiver divides received frame by **same polynomial**.
5. If remainder = **0 → No error**.

Example Polynomials

- CRC-8
- CRC-12
- CRC-16
- CRC-32

Best for:

Burst errors, long data transmission.

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3. ERROR CORRECTION TECHNIQUES

Types of Error Correction

3.1 Forward Error Correction (FEC)

- Errors are corrected **without retransmission**.
- Sender adds **redundant bits**.
- Used in: satellite, live audio/video, wireless.

3.2 Backward Error Correction (ARQ)

- Sender retransmits data if receiver finds errors.
- ARQ = Automatic Repeat reQuest.

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4. DATA LINK CONTROL

Data Link Control refers to all techniques to ensure **reliable communication**.

Main components:

1. **Framing**
2. **Flow Control**
3. **Error Control**
4. **ARQ Protocols**

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5. FRAMING

Framing divides the data stream into **manageable units called frames**.

Methods:

5.1 Character Count

- First field of frame = number of characters.
- Problem: If count field is corrupted → entire frame lost.

5.2 Flag Bytes with Byte Stuffing

- Uses special **flag byte** → 01111110 (0x7E)
- If data contains same byte → insert escape byte **(ESC)**

Example:

Data: 7E
Stuffing → ESC + 7E

5.3 Bit Stuffing

- Used in HDLC.
- Flag is → 01111110
- After 5 consecutive 1s in data → insert a 0.

Example:
Data: 111110
Transmitter adds → 0 after 5 ones.

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6. FLOW CONTROL

Ensures sender does NOT overwhelm receiver.

6.1 Stop-and-Wait Flow Control

- Sender → sends 1 frame
- Waits for **ACK** before next frame

Simple but slow.

6.2 Sliding Window Flow Control

- Sender can transmit **multiple frames** based on window size.
- Receiver also has a window.

Used in: TCP, ARQ protocols.

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7. ERROR RECOVERY PROTOCOLS (ARQ)

ARQ = Automatic Repeat reQuest
Used for **error control + reliable delivery**.

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8. STOP-AND-WAIT ARQ

Used for both **flow control + error control**.

Working:

1. Sender sends one frame.
2. Waits for ACK (acknowledgement).
3. If ACK lost or damaged → retransmits frame.
4. Uses **sequence numbers 0 and 1**.

Problems:

- Slow for long-distance communication.
- Low efficiency.

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9. GO-BACK-N ARQ

Uses **Sliding Window**.

Window Sizes:

- Sender Window Size = **N**
- Receiver Window Size = **1**

Working:

1. Sender sends multiple frames (0...N).
2. If an error occurs in any frame:
   → Receiver discards that frame and **all following frames**
   → Receiver sends **NACK**
   → Sender retransmits **from the error frame onward ("Go Back")**.

Example:

Sent: 0,1,2,3,4
Error in 2 → Receiver discards 2,3,4
Sender retransmits 2,3,4.

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10. SELECTIVE REPEAT ARQ *(Mention for clarity)*

(Not required in your syllabus but useful to know.)

- Receiver window > 1
- Only incorrect frames are retransmitted.

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11. POINT-TO-POINT PROTOCOL (PPP)

Used for **point-to-point** communication such as:

- WAN links
- Dial-up connections
- DSL broadband
- VPN tunnels

PPP Frame Format

| Flag | Address | Control | Protocol | Payload | FCS | Flag |

Fields:

1. **Flag (01111110)** – start & end of frame
2. **Address** – always 11111111 (broadcast)
3. **Control** – usually 00000011
4. **Protocol** – identifies type of data (IP, LCP, NCP)
5. **Payload** – actual data
6. **FCS** – checksum/CRC
7. **Flag** – marks end

PPP Components

1. **LCP – Link Control Protocol**
    - Establish link
    - Configure link
    - Authenticate (PAP/CHAP)
    - Terminate link
2. **NCP – Network Control Protocol**
    - Allows using different Network Layer protocols
    - Example: IPCP for IP

PPP Advantages

- Simple
- Supports error detection
- Supports authentication
- Works with many network layer protocols
- Uses HDLC-like framing

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12. SUMMARY NOTES (for revision)

Error Detection:

- Parity
- Checksum
- CRC (best)

Error Correction:

- FEC (no retransmission)
- ARQ (retransmission)

Framing:

- Character Count
- Byte Stuffing
- Bit Stuffing

Flow Control:

- Stop and Wait
- Sliding Window

ARQ Protocols:

- Stop-and-Wait ARQ
- Go-Back-N ARQ
- (Selective Repeat optional)

PPP:

- WAN protocol
- Uses LCP + NCP
- Uses bit-stuffing
- HDLC-style framing